

AQA A-Level Further Mathematics (7367)

Question Paper

Further Calculus

E2 - Volume Revolution

Name: _____

Class: _____

Date: _____

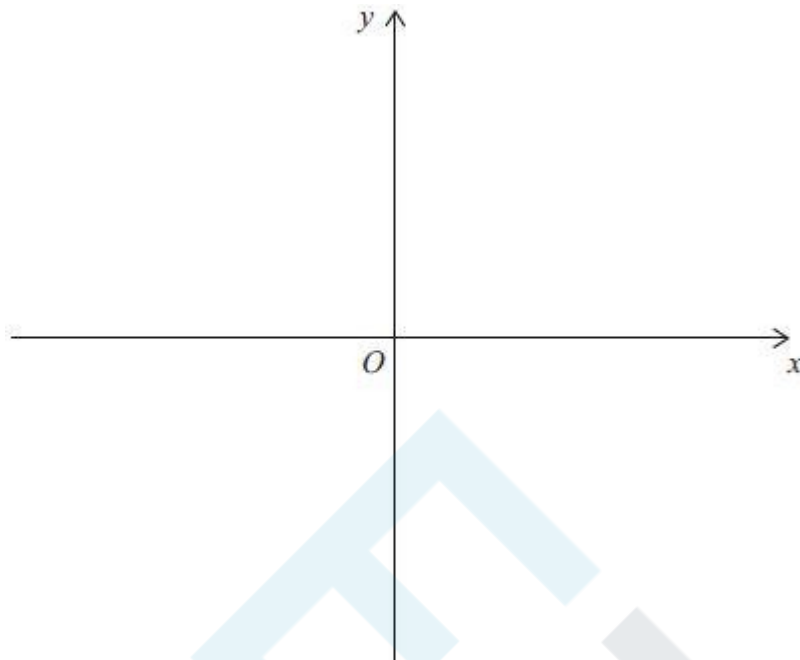
Time: 195 min.

Marks: 162 marks

Comments:

Q1.

- (a) Sketch the graph of $y^2 = 4x$



(1)

- (b) Ben is using a 3D printer to make a plastic bowl which holds exactly 1000 cm^3 of water.

Ben models the bowl as a region which is rotated through 2π radians about the x -axis.

He uses the finite region enclosed by the lines $x = d$ and $y = 0$ and the curve with equation $y^2 = 4x$ for $y \geq 0$

- (i) Find the depth of the bowl to the nearest millimetre.

(4)

- (ii) What assumption has Ben made about the bowl?

(1)

(Total 6 marks)

Q2.

The region bounded by the curve with equation $y = 3 + \sqrt{x}$, the x -axis and the lines $x = 1$ and $x = 4$ is rotated through 2π radians about the x -axis.

$$\frac{125\pi}{2}$$

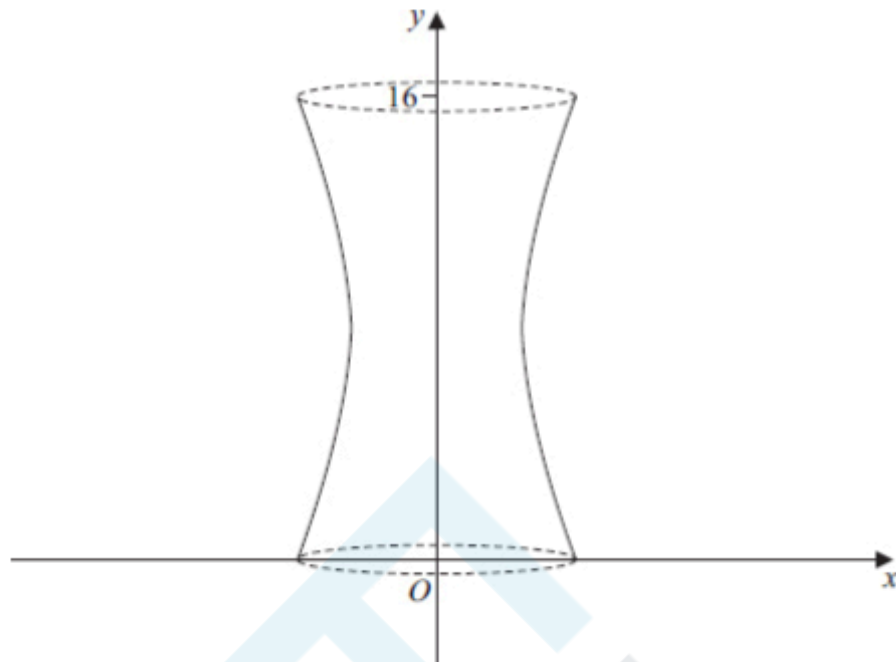
Use integration to show that the volume generated is

(Total 5 marks)

Q3.



The shape of a vase can be modelled by rotating the curve with equation $16x^2 - (y - 8)^2 = 32$ between $y = 0$ and $y = 16$ completely **about the y -axis**.



The vase has a base.

Find the volume of water needed to fill the vase, giving your answer as an exact value.

(Total 5 marks)

Q4.

- (a) A curve has equation

$$y = x^2 e^{-\frac{x}{4}}.$$

Show that the curve has exactly two stationary points and find the exact values of their coordinates.

(7)

- (b) (i) Use integration by parts twice to find the exact value of $\int_0^4 x^2 e^{-\frac{x}{4}} dx$.

(7)

- (ii) The region bounded by the curve $y = 3xe^{-\frac{x}{8}}$, the x -axis from 0 to 4 and the line $x = 4$ is rotated through 360° about the x -axis to form a solid.

Use your answer to part (b)(i) to find the exact value of the volume of the solid generated.

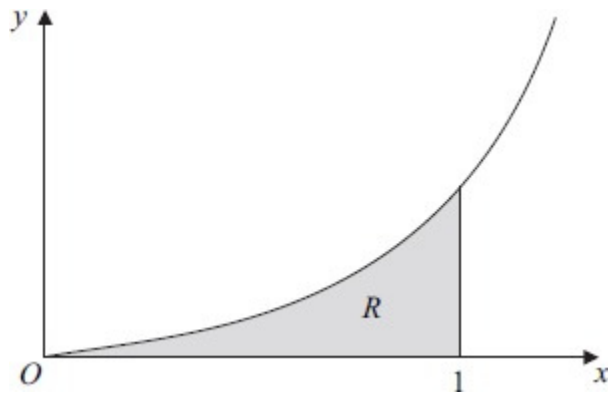
(2)

(Total 16 marks)

Q5.

- (a) By using integration by parts, find $\int x e^{6x} dx$.

- (b) The diagram shows part of the curve with equation $y = \sqrt{x} e^{3x}$.



The shaded region R is bounded by the curve $y = \sqrt{x} e^{3x}$, the line $x = 1$ and the x -axis from $x = 0$ to $x = 1$.

Find the volume of the solid generated when the region R is rotated through

360°

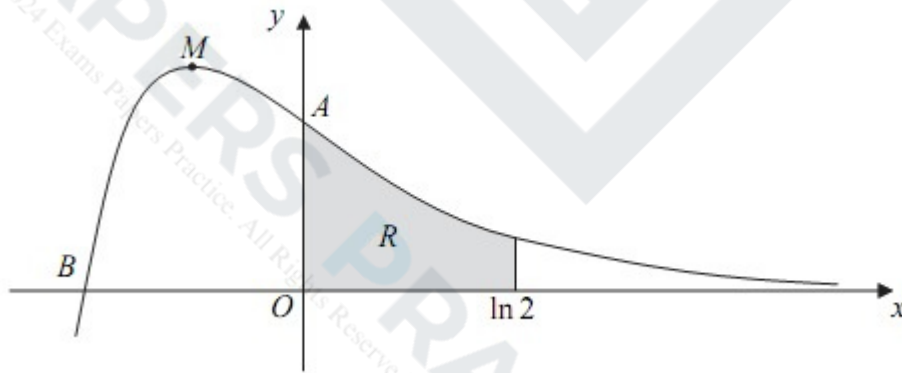
about the x -axis, giving your answer in the form $\pi(p e^6 + q)$, where p and q are rational numbers.

(3)

(Total 7 marks)

Q6.

The diagram shows the curve $y = 4e^{-2x} - e^{-4x}$.



The curve crosses the y -axis at the point A , the x -axis at the point B , and has a stationary point at M .

The shaded region R is bounded by the curve $y = 4e^{-2x} - e^{-4x}$, the lines $x = 0$ and $x = \ln 2$ and the x -axis.

Find the volume of the solid generated when the region R is rotated through

360° about the x -axis, giving your answer in the form $\frac{p}{q}\pi$, where p and q are integers.

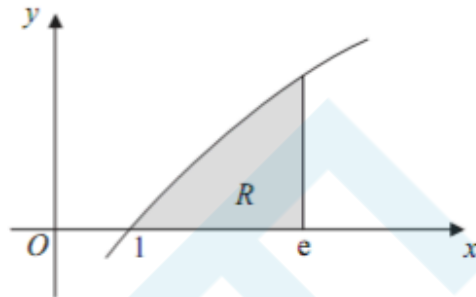
(Total 7 marks)

Q7.

- (a) Use integration by parts to find $\int x \ln x \, dx$. (3)

- (b) Given that $y = (\ln x)^2$, find $\frac{dy}{dx}$. (2)

- (c) The diagram shows part of the curve with equation $y = \sqrt{x} \ln x$.



The shaded region R is bounded by the curve $y = \sqrt{x} \ln x$, the line $x = e$ and the x -axis from $x = 1$ to $x = e$.

Find the volume of the solid generated when the region R is rotated through 360° about the x -axis, giving your answer in an exact form.

(6)

(Total 11 marks)

Q8.

- (a) Use the mid-ordinate rule with four strips to find an estimate for $\int_0^{12} \ln(x^2 + 5) \, dx$, giving your answer to three significant figures. (4)

- (b) A curve has equation $y = \ln(x^2 + 5)$.

- (i) Show that this equation can be rewritten as $x^2 = e^y - 5$. (1)

- (ii) The region bounded by the curve, the lines $y = 5$ and $y = 10$ and the y -axis is rotated through 360° about the y -axis. Find the exact value of the volume of the solid generated.

(4)

(Total 9 marks)

Q9.

- (a) Use integration by parts to find:

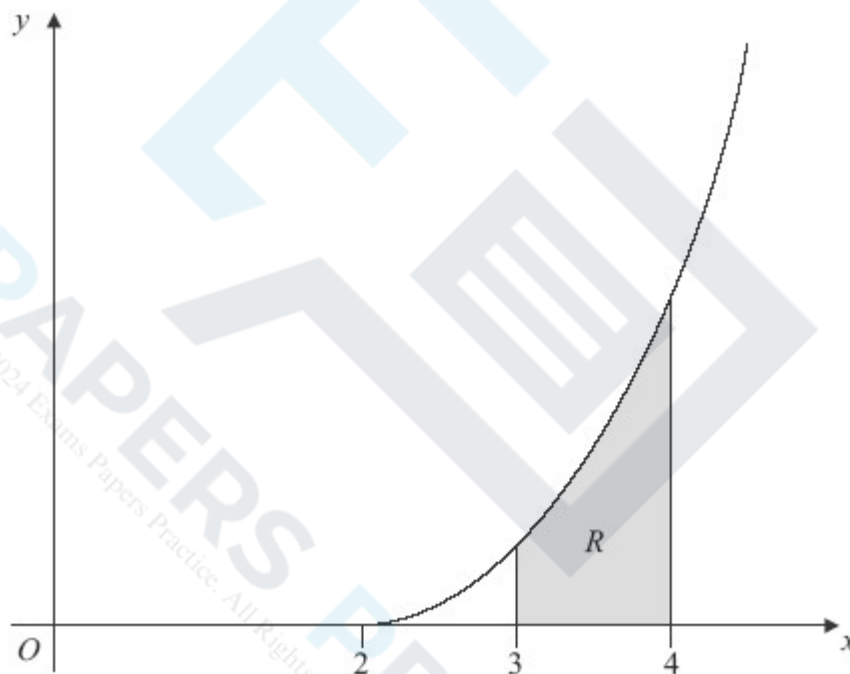
(i) $\int x \cos 4x \, dx$; (4)

(ii) $\int x^2 \sin 4x \, dx$. (4)

- (b) The region bounded by the curve $y = 8x\sqrt{\sin 4x}$ and the lines $x = 0$ and $x = 0.2$ is rotated through 2π radians about the x -axis. Find the value of the volume of the solid generated, giving your answer to three significant figures. (3)
- (Total 11 marks)

Q10.

The diagram shows the curve with equation $y = \sqrt{(x-2)^5}$ for $x \geq 2$.



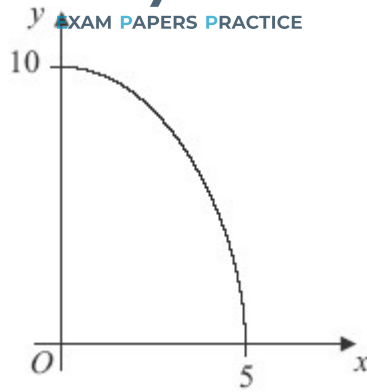
The shaded region R is bounded by the curve $y = \sqrt{(x-2)^5}$, the x -axis and the lines $x = 3$ and $x = 4$.

Find the exact value of the volume of the solid formed when the region R is rotated through 360° about the x -axis.

(Total 4 marks)

Q11.

The diagram shows the curve with equation $y = \sqrt{100 - 4x^2}$, where $x \geq 0$.



- (a) Calculate the volume of the solid generated when the region bounded by the curve shown above and the coordinate axes is rotated through 360° about the y-axis, giving your answer in terms of π . (5)
- (b) Use the mid-ordinate rule with five strips of equal width to find an estimate for $\int_0^5 \sqrt{100 - 4x^2} \, dx$, giving your answer to three significant figures. (4)
- (Total 9 marks)

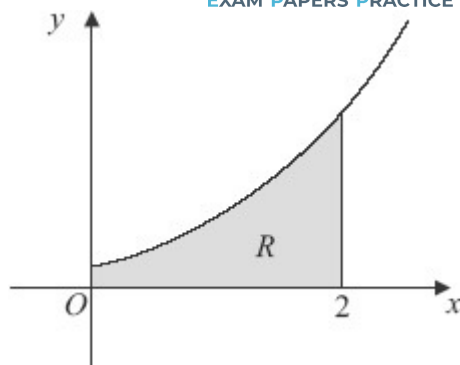
Q12.

- (a) Given that $e^{-2x} = 3$, find the exact value of x . (2)
- (b) Use integration by parts to find $\int x e^{-2x} \, dx$. (4)
- (c) A curve has equation $y = e^{-2x} + 6x$.
- (i) Find the exact values of the coordinates of the stationary point of the curve. (4)
- (ii) Determine the nature of the stationary point. (2)
- (iii) The region R is bounded by the curve $y = e^{-2x} + 6x$, the x-axis and the lines $x = 0$ and $x = 1$. Find the volume of the solid formed when R is rotated through 2π radians about the x-axis, giving your answer to three significant figures. (5)

(Total 17 marks)

Q13.

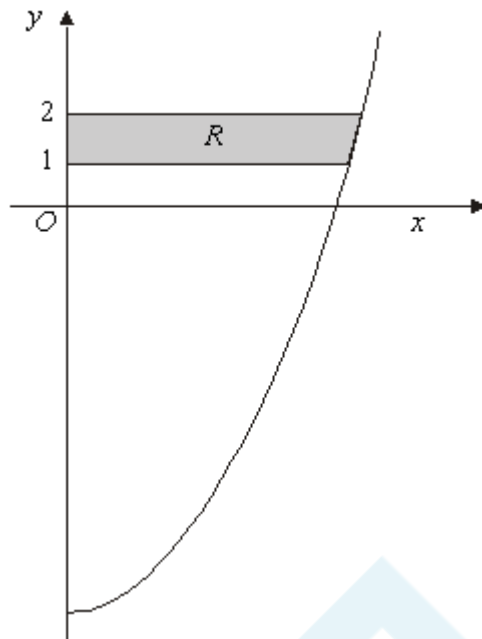
The diagram shows the curve with equation $y = (e^{3x} + 1)^{\frac{1}{2}}$ for $x \geq 0$.



- (a) Find the gradient of the curve $y = (e^{3x} + 1)^{\frac{1}{2}}$ at the point where $x = \ln 2$. (5)
- (b) Use the mid-ordinate rule with four strips to find an estimate for $\int_0^2 (e^{3x} + 1)^{\frac{1}{2}} dx$, giving your answer to three significant figures. (4)
- (c) The shaded region R is bounded by the curve, the lines $x = 0$, $x = 2$ and the x -axis. Find the exact value of the volume of the solid generated when the region R is rotated through 360° about the x -axis. (4)
- (Total 13 marks)

Q14.

- (a) Use integration by parts to find $\int x \sin x \, dx$. (4)
- (b) Using the substitution $u = x^2 + 5$, or otherwise, find $\int x \sqrt{x^2 + 5} \, dx$. (4)
- (c) The diagram shows the curve $y = x^2 - 9$ for $x \geq 0$.



The shaded region R is bounded by the curve, the lines $y = 1$ and $y = 2$, and the y -axis.
Find the exact value of the volume of the solid generated when the region R is rotated through 360° about the y -axis.

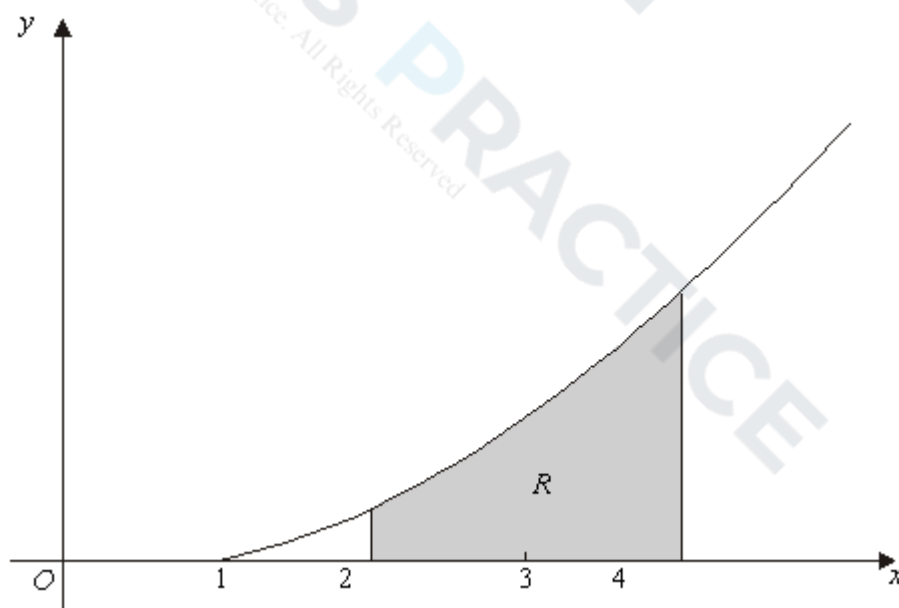
(4)
(Total 12 marks)

Q15.

- (a) Differentiate $(x-1)^4$ with respect to x .

(1)

- (b) The diagram shows the curve with equation $y = 2\sqrt{(x-1)^3}$ for $x \geq 1$.



The shaded region R is bounded by the curve $y = 2\sqrt{(x-1)^3}$, the lines $x = 2$ and $x = 4$, and the x -axis.

Find the exact value of the volume of the solid formed when the region R is rotated through 360° about the x -axis.

(4)

- (c) Describe a sequence of two geometrical transformations that maps the graph of $y = \sqrt{x^3}$ onto the graph of $y = 2\sqrt{(x-1)^3}$

(4)

(Total 9 marks)

Q16.

- (a) Given that $z = \frac{\sin x}{\cos x}$, use the quotient rule to show that $\frac{dz}{dx} = \sec 2x$.

(3)

- (b) Sketch the curve with equation $y = \sec x$ for $\frac{\pi}{2} < x < \frac{\pi}{2}$.

(2)

- (c) The region R is bounded by the curve $y = \sec x$, the x -axis and the lines $x = 0$ and $x = 1$.

Find the volume of the solid formed when R is rotated through 2π radians about the x -axis, giving your answer to three significant figures.

(3)

(Total 8 marks)

Q17.

- (a) (i) Sketch the graph of $y = 4 - x^2$, indicating the coordinates of the points where the graph crosses the coordinate axes.

(2)

- (ii) The region between the graph and the x -axis from $x = 0$ to $x = 2$ is rotated through 360° about the x -axis. Find the exact value of the volume of the solid generated.

(4)

- (b) (i) Sketch the graph of $y = |4 - x^2|$.

(2)

- (ii) Solve $|4 - x^2| = 3$.

(3)

- (iii) Hence, or otherwise, solve the inequality $|4 - x^2| < 3$.

(2)

(Total 13 marks)